Exercise-3

H2C, IOT Project

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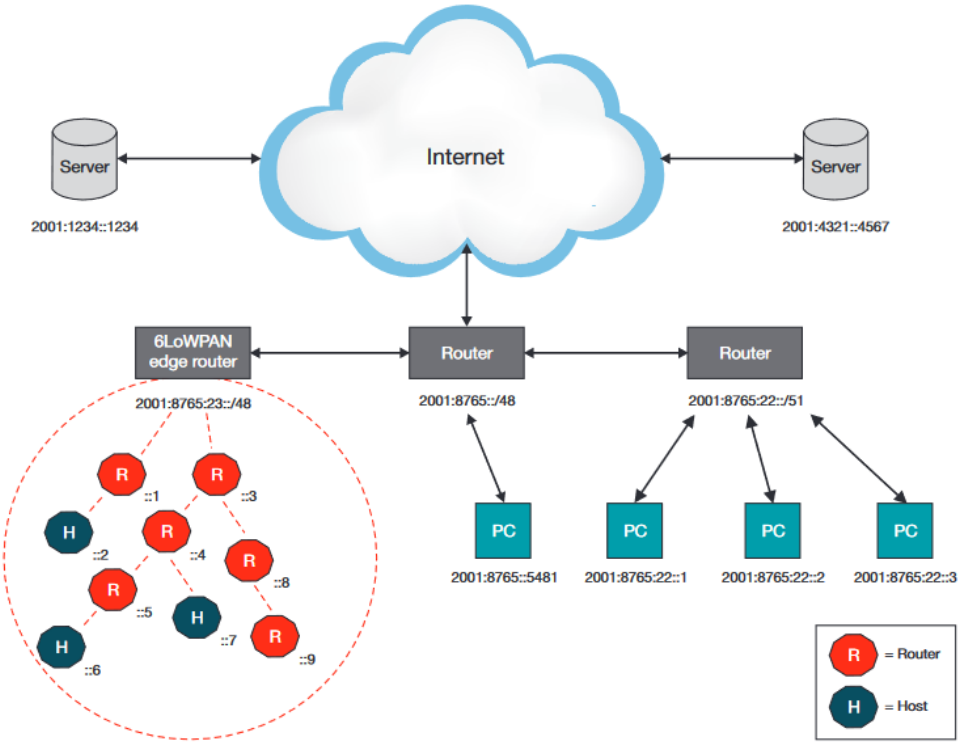
1.1 6LoWPAN

2. Task-2

2.1. Routing Challenges in the Internet of Things

6LoWPAN

6LoWPAN stands for 'IPv6 over Low-Power Wireless Personal Area Networks'. This system is used for a different kind of applications with wireless sensor networks. Low power wireless mesh network is a powerful feature of 6LoWPAN, and each node included its own IPv6 address. This help all nodes to connect directly with the main Internet using open standards. 6LoWPAN was the idea that the IP could and should be used to the smallest devices, and that could be applied in the low-power devices and will be able to participate in the Internet of Things. 6LoWPAN provides to carry packet data in the form of IPv6 over IEEE 802.15.4 and other networks. It provides end-to-end IPv6 infrastructure. Thus, 6LoWPAN helps to provide a different low power wireless sensor network solution. (Jonas Olsson)



An example of an IPv6 network with a 6LoWPAN mesh network. (Adapted from Jonas Olsson)

6LoWPAN works with open IP standard including TCP, UDP, HTTP, COAP, MATT and web-sockets. It provides end-to-end IP addressable nodes in scalable mesh network. That is why a gateway is not needed, only a router can connect the 6LoWPAN network to IP. It offers one-to-many and many-to-one routing. The 6LoWPAN mesh routers can route data to other nodes over the network. In a 6LowPAN network, one node can sleep for a long duration of time, that

will not affect other nodes to run. It also offers connectivity to others IP networks through arbitrary link like Ethernet, WI-Fi, 802.15.4 or Sub-1GHz ISM. Security is a big issue for any standard like 6LoWPAN, AES-128 link layer is used for security which is defined in IEEE 802.15.4. It provides link of authentication and encryption. Furthermore, security is provided by the transport layer security system which is also included. This is defined in RFC 5246 and runs over TCP. Transport layer protocol what is defined under RFC 6347 is used by the UDP, though it may require some hardware requirements. (Jonas Olsson)

It is very important to define a new network with the addresses when there are a lot of low power wireless sensor networks. It is needed for Embedded devices to communicate with Internet-based services and low-power various networks need to be attached together. Connecting the network to the physical world is very interesting applications where 6LoWPAN technology is needed. However, 6LoWPAN is being used in many applications, for example in general automation, home automation, industrial IoT, smart agriculture and healthcare automation. There has some drawbacks of 6LoWPAN, such as less security, supports only short range without mesh topology. (EETimes. 2011.)

Routing Challenges in the Internet of Things

Firstly, there is a discussion about IoT goals, such extensive interconnection, more intensive information perception and more comprehensive intelligence service. Because of the short range of networks, devices are connected to the wireless sensor networks (WSNs), radio frequency identification (RFID) networks, Bluetooth and Zigbee networks. The article is about the types of routing protocol, routing challenges that need to be addressed. Routing protocols are divided into proactive, reactive and hybrid routing protocols. Proactive protocols always maintain the route information in tabular format at any time, reactive protocols builds the on-demand route which is suitable in WSN, whereas hybrid routing makes use of both proactive and reactive routing algorithms. There are some big challenges may arise in the routing process of IoT, such as deployment of nodes, heterogeneous devices, multihop communication, fault tolerance security and context awareness. Finally, it is important that every possible ‘thing’ need to connect to the network with the addressing routing issues.

EETimes. 2011. 6LoWPAN: The wireless embedded Internet – Part 2: 6LoWPAN history, market perspective & applications. Available: https://www.eetimes.com/6lowpan-the-wireless-embedded-internet-part-2-6lowpan-history-market-perspective-applications/

Jonas Olsson. 6LoWPAN demystified